

catered to by brands such as Shure, Beyerdynamic, AKG, and others. Planar magnetic technology and electrostatic technology headphones were now readily available, so this was quite a great time for audio purists as well.

## THE ADVENT OF WIRELESS AUDIO AND WHERE WE ARE TODAY

A century after the invention of headphones, they went wireless due to the development of Bluetooth technology. This is arguably, one of the biggest leaps in headphone innovation, if not the biggest. Bluetooth headphones made their grand entry into the headphone scene in the mid-2000s, although it wasn't

from wireless headphones for a long time since the technology simply wasn't robust enough to provide results close to or matching that of their wired counterparts. Yet, vast research in the field kept going on and lossless codecs as well as hi-resolution codecs such as Qualcomm's aptX HD, Sony's LDAC, Apple's AAC, Savitech's LHDC, and others were developed. These codecs claim CD-like sound quality and the general reception has been overwhelmingly positive. Nevertheless, Bluetooth earphones and headphones gained more and more momentum and they're pretty much ubiquitous these days.

One of the biggest steps forward in the last decade in headphone inno-

7, and the Apple AirPods (2016) did catapult this category, which was in its infancy, to the limelight.

What followed was a flurry of activity by audio brands jumping on the new, lucrative trend with brands coming out with their own renditions of truly wireless earphones. As the years went by, companies were able to achieve more and more in this compact form factor by improving on the core design and functionality of early TWS (truly wireless) earphones. Today, we have TWS earphones with larger batteries, bigger drivers, multiple drivers, Active Noise Cancellation technology, and much more. It would be almost inconceivable to the early makers of headphones that these specs and features could be incorporated into such a diminutive form factor! Moreover, now you also see the addition of innovative features such as wear detection (auto pause/play), voice assistant support, noise cancelling based on the environment, and others. While TWS earphones are certainly what's new and exciting, headphone form factors such as on-ear headphones, over-ear headphones, and neckband-style headphones (both wired and wireless) are also seeing the incorporation of some of the innovative technologies we mentioned such as ANC, wear detection, voice assistant support, and more. Gaming headsets also have a sizable market and are also constantly getting better with innovations such as surround sound, head tracking, 3D audio, microphone ambient noise cancellation, and more.



Onkyo W800 BT, the world's first truly wireless earphones

so grand, since they only came into the limelight in the early 2010s. Soon after, earphones or in-ear headphones also followed suit and became wireless like their on-ear or over-ear counterparts. Audio brands quickly saw the opportunity to leverage this boom and the brands, regardless of whether they were commercial brands or audiophile ones, jumped in and developed their own wireless models.

Nevertheless, a vast number of audiophiles or purists swayed away

from the creation of truly wireless in-ear headphones. Contrary to popular belief, it wasn't Apple that launched the world's first truly wireless earphone, but a Japanese brand called Onkyo. The company unveiled the world's first fully untethered pair of earbuds, dubbed the W800 BT at the IFA Show in Berlin in 2015. But credit must be given where it's due, since Apple did have a massive part in bringing about the truly wireless revolution by removing the headphone jack on the iPhone

## THE WAY FORWARD

The headphone industry witnessed a paradigm shift since the launch of the AirPods in 2018. Growth and innovation in the industry is expected to continue at lightning speed. As per the Earphones and Headphones Market - Global Outlook and Forecast 2021-2026 by ReportLinker, the industry is touted to grow at a CAGR of over 13 percent during 2019-2025. A recent Spotify study confirmed that a majority of millen-